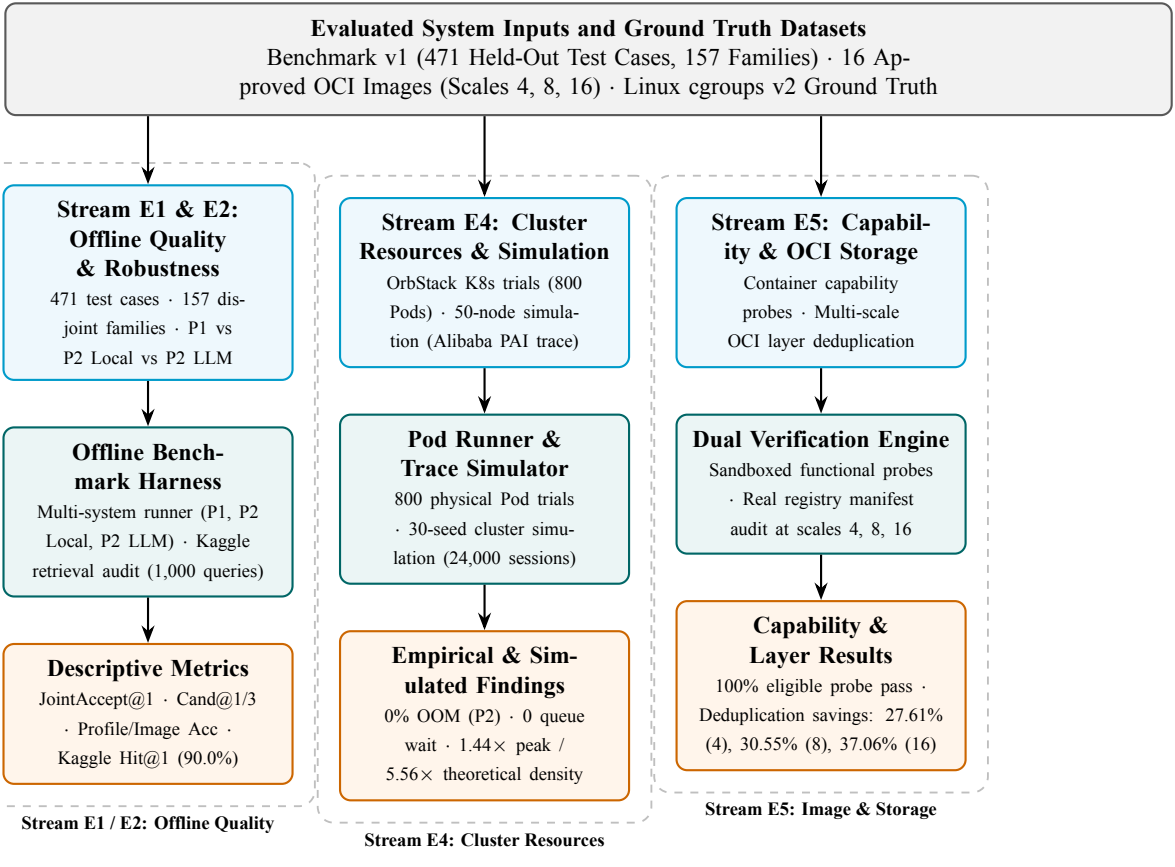

Evaluated System Inputs and Ground Truth Datasets
Benchmark v1 (471 Held-Out Test Cases, 157 Families) · 16 Approved OCI Images (Scales 4, 8, 16) · Linux cgroups v2 Ground Truth
Stream E1 & E2: Offline Quality & Robustness
471 test cases · 157 disjoint families · P1 vs P2 Local vs P2 LLM
Offline Benchmark Harness
Multi-system runner (P1, P2 Local, P2 LLM) · Kaggle retrieval audit (1,000 queries)
Descriptive Metrics
JointAccept@1 · Cand@1/3 · Profile/Image Acc · Kaggle Hit@1 (90.0%)
Stream E1 / E2: Offline Quality
Stream E4: Cluster Resources & Simulation
OrbStack K8s trials (800 Pods) · 50-node simulation (Alibaba PAI trace)
Pod Runner & Trace Simulator
800 physical Pod trials · 30-seed cluster simulation (24,000 sessions)
Empirical & Simulated Findings
0% OOM (P2) · 0 queue wait · 1.44× peak / 5.56× theoretical density
Stream E4: Cluster Resources
Stream E5: Capability & OCI Storage
Container capability probes · Multi-scale OCI layer deduplication
Dual Verification Engine
Sandboxed functional probes · Real registry manifest audit at scales 4, 8, 16
Capability & Layer Results
100% eligible probe pass · Deduplication savings: 27.61% (4), 30.55% (8), 37.06% (16)
Stream E5: Image & Storage